Vectors in n Dimensions

Vector addition and scalar multiplication

1 Let $\mathbf{u} = \langle 2, -1, 3\rangle$, $\mathbf{v} = \langle 1, 4, -2\rangle$. Find $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.

Dot product and norm

2 For $\mathbf{u} = \langle 1, 2, 2\rangle$, $\mathbf{v} = \langle 3, -1, 0\rangle$, find $\mathbf{u} \cdot \mathbf{v}$ and $|\mathbf{u}|$.

3 Find the angle between $\mathbf{u} = \langle 1, 0, 0\rangle$ and $\mathbf{v} = \langle 1, 1, 0\rangle$ using $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

The Cauchy–Schwarz Inequality

4 State the Cauchy–Schwarz Inequality.

5 Verify the Cauchy–Schwarz Inequality for $\mathbf{u} = \langle 3, 4\rangle$, $\mathbf{v} = \langle 1, 2\rangle$.

Unit vectors

6 Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4\rangle$.

Orthogonality

7 Determine whether $\mathbf{u} = \langle 2, -3, 1\rangle$ and $\mathbf{v} = \langle 1, 2, 4\rangle$ are orthogonal.

8 For $\mathbf{u} = \langle 1, -1, 2\rangle$, find $|\mathbf{u}|$ and the unit vector $\hat{\mathbf{u}}$.

More vector arithmetic

9 Let $\mathbf{u} = \langle 0, 3, -1\rangle$, $\mathbf{v} = \langle 4, -2, 5\rangle$. Find $2\mathbf{u} + 3\mathbf{v}$ and $\mathbf{u} - \mathbf{v}$.

More dot product and norm practice

10 For $\mathbf{u} = \langle 2, 2, 1\rangle$, $\mathbf{v} = \langle 0, 3, 4\rangle$, find $\mathbf{u} \cdot \mathbf{v}$, $|\mathbf{u}|$, $|\mathbf{v}|$.

More angle-between-vectors practice

11 Find the angle between $\mathbf{u} = \langle 1, 1, 0\rangle$ and $\mathbf{v} = \langle 0, 1, 1\rangle$.

The triangle inequality

12 State the triangle inequality for vectors.

13 Verify the triangle inequality for $\mathbf{u} = \langle 3, 0\rangle$, $\mathbf{v} = \langle 0, 4\rangle$.

Verifying a unit vector

14 For $\mathbf{u} = \langle 1, 2, -2\rangle$, find $|\mathbf{u}|$, and verify that $\hat{\mathbf{u}} = \mathbf{u}/|\mathbf{u}|$ satisfies $|\hat{\mathbf{u}}| = 1$.

Vector projection

15 Find the projection of $\mathbf{u} = \langle 3, 4\rangle$ onto $\mathbf{v} = \langle 1, 0\rangle$, using $\text{proj}_{\mathbf{v}}\mathbf{u} = \dfrac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v}$.

16 Find the projection of $\mathbf{u} = \langle 2, 3\rangle$ onto $\mathbf{v} = \langle 1, 1\rangle$.

Distance between two points as a vector norm

17 Find the distance between $P(1, 2, 3)$ and $Q(4, -1, 5)$ using $|\overrightarrow{PQ}|$.

Linear combinations of vectors

18 Write $\mathbf{w} = \langle 7, 4\rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 0\rangle$ and $\mathbf{v} = \langle 0, 1\rangle$.

19 Write $\mathbf{w} = \langle 5, 1\rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 1\rangle$ and $\mathbf{v} = \langle 1, -1\rangle$.

Vectors from two points

20 Find the vector $\overrightarrow{PQ}$ from $P(2, -1, 0)$ to $Q(5, 3, -2)$.

21 Find $|\overrightarrow{PQ}|$ for the vector found above.

Scalar multiples and direction

22 For $\mathbf{v} = \langle 2, -3\rangle$, is $-4\mathbf{v}$ in the same direction as $\mathbf{v}$, or the opposite?

23 Are $\mathbf{u} = \langle 2, -6\rangle$ and $\mathbf{v} = \langle -1, 3\rangle$ parallel?

The midpoint of a segment as a vector average

24 Find the midpoint M of the segment from $P(1, -2, 3)$ to $Q(5, 4, -1)$, using $M = \frac{1}{2}(P + Q)$ (averaging componentwise).

Normalizing a vector to a given length

25 Find the vector of length 10 in the same direction as $\mathbf{v} = \langle 3, 4 \rangle$.